

2.04 Statistical Distributions				
2.04a	Discrete probability distributions	a) Understand and be able to use simple, finite, discrete probability distributions, defined in the form of a table or a formula such as: $P(X = x) = 0.05x(x + 1)$ for $x = 1, 2, 3$. <i>[Calculation of mean and variance of discrete random variables is excluded.]</i>		MN1 MN2 MN3
2.04b		b) Understand and be able to use the binomial distribution as a model.		
2.04d		c) Be able to calculate probabilities using the binomial distribution, using appropriate calculator functions.		
2.04c		<i>Includes understanding and being able to use the formula</i> $P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x} \text{ and the notation } X \sim B(n, p).$ <i>Learners should understand the conditions for a random variable to have a binomial distribution, be able to identify which of the modelling conditions (assumptions) is/are relevant to a given scenario and be able to explain them in context. They should understand the distinction between conditions and assumptions.</i>		
		d) Know and be able to use the formulae $\mu = np$ and $\sigma^2 = npq$ when choosing a particular normal model to use as an approximation to a binomial model.		

OCR Ref.	Subject Content	Stage 1 learners should ...	Stage 2 learners additionally should ...	DfE Ref.
2.04e 2.04f 2.04g	The normal distribution		e) Understand and be able to use the normal distribution as a model. <i>Includes understanding and being able to use the notation $X \sim N(\mu, \sigma^2)$.</i> f) Be able to find probabilities using the normal distribution, using appropriate calculator functions. <i>This includes finding x, for a given normal variable, when $P(X < x)$ is known.</i> <i>Learners should understand the standard normal distribution, Z, and the transformation $Z = \frac{X - \mu}{\sigma}$.</i> g) Understand links to histograms, mean and standard deviation. <i>Learners should know and be able to use the facts that in a normal distribution,</i> <i>1. about two-thirds of values lie in the range $\mu \pm \sigma$,</i> <i>2. about 95% of values lie in the range $\mu \pm 2\sigma$,</i> <i>3. almost all values lie in the range $\mu \pm 3\sigma$ and</i> <i>4. the points of inflection in a normal curve occur at $x = \mu \pm \sigma$.</i> <i>[The equation of the normal curve is excluded.]</i>	MN2
2.04h	Selecting an appropriate distribution		h) Be able to select an appropriate probability distribution for a context, with appropriate reasoning, including recognising when the binomial or normal model may not be appropriate. <i>Includes understanding that a given binomial distribution with large n can be approximated by a normal distribution.</i> <i>[Questions explicitly requiring calculations using the normal approximation to the binomial distribution are excluded.]</i>	MN2 MN3

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